

Positioning performance analysis of real-time BDS-3 PPP-B2b/INS tightly coupled integration in urban environments

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Abstract

Since 2020, the BeiDou-3 global navigation satellite system has provided real-time precision single-point positioning enhancement services (PPP-B2b). The real-time correction provided by this service can make real-time precision single-point positioning achieve decimeter-level. However, a dense urban environment may cause frequent interruption of GNSS signals, and PPP-B2b alone cannot achieve continuous precision positioning. This study documents a study that attempts to combine PPP-B2b with an inertial navigation system (INS). A real-time tightly coupled integration model of PPP-B2b/INS based on PPP-B2b services was constructed, and a performance analysis of the tightly coupled integration positioning of PPP-B2b/INS in different urban environments was performed. The experiment shows that in an open-sky environment, the decimeter-level positioning accuracy (North: 31.7; East: 21.3; Up: 39.6 cm and North: 34.1; East: 27.6; Up: 41.0 cm) can only be achieved based on PPP-B2b (BDS only and GPS only), PPP-B2b/INS tightly coupled integration positioning accuracy (North: 17.2; East: 10.3; Up: 19.4 cm), and the convergence time is faster; In a dense urban environment, GNSS signals are frequently interrupted, and the positioning accuracy (North: 141.3; East: 95.2; Up: 156.6 cm and North: 147.5; East: 96.4; Up: 159.3 cm) increases significantly only based on PPP-B2b (BDS only and GPS only), and positioning interruption occurs frequently. However, PPP-B2b/INS tightly coupled integration could still achieve continuous decimetre-level positioning (North: 30.0; East: 28.4; Up: 46.3 cm).

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1. Introduction

With the construction and gradual improvement of global satellite navigation systems (GPS, GLONASS, Galileo, and BDS), precision point positioning (PPP) is widely used (Zumberge et al., 1997), especially in the fields of automatic driving and intelligent navigation. The demand for high-

precision real-time positioning is increasingly urgent (Chen and Chang, 2020). In recent years, with precision products using real-time service (RTS) transmission provided by Internet communication through the Internet GNSS service (IGS) (Hadas and Bosy, 2015; Héroux and Kouba, 2001), users can accurately calculate the satellite orbit and clock offset to achieve real-time PPP, which has become a widely used high-precision navigation and positioning method. However, RTS is interrupted when the Internet is unavailable, which is fatal to RTS-based real-time PPP (Angrisano et al., 2013; Gong et al., 2020). Since 2020, the satellite-based augmentation system (SBAS) of

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the Beidou-3 navigation satellite system (BDS-3) has broadcasted the precise orbit, clock offset, intersymbol deviation, and other correction parameters of BDS-3 and the Global Positioning System (GPS) in real time through GEO, effectively solving the problem that real-time PPP cannot be implemented when the Internet is interrupted (Ren et al., 2020; Yang et al., 2021; Liu et al., 2022). Owing to the fact that the BDS-3 PPP-B2b service is based on satellites and does not require Internet communication, the PPP-B2b service has great potential for real-time global navigation and positioning systems. The corrections provided by PPP-B2b service can meet the requirements of high-precision positioning in China and surrounding areas in terms of time and regional coverage, and the accuracy of the correction is similar to that of precision products released by international monitoring stations (Yang et al., 2022).

Generally, GNSS signal quality is good in an open-sky environment, and precision positioning at the decimetre level can be achieved using the PPP-B2b service (Zhang et al., 2022). However, the GNSS signal is frequently blocked or interfered within a dense urban environment, and the interruption of the GNSS signal leads to frequent reconvergence of the PPP, which cannot obtain continuous precision positioning (Collins et al., 2010). INS features high frequency, high precision (Zhang and Gao, 2008; Li et al., 2017), and continuous positioning, so continuous high-precision positioning can still be obtained after the GNSS signal is interrupted, and the PPP/INS tightly coupled integration model can largely solve the problems caused by the number of satellites (Gao et al., 2017; Gu et al., 2021), especially in urban environments that are often sheltered by tall buildings. When the number of visible satellites is less than 4 but not less than 1, the tightly coupled integration model can still perform measurement update and limit error accumulation (Gao et al., 2015). At the same time, in the application of the PPP/INS tightly coupled integration model, precision products of the IGS real-time service are generally used for navigation solutions, but the IGS RTS service is often limited (Hadas and Bosy, 2015; Nie et al., 2018), owing to the acquisition time of final precision products and Internet connection; thus, the application of real-time corrections provided by the BDS-3 PPP-B2b service is also of great significance for PPP/INS tightly coupled integration.

At present, the published PPP-B2b related research mainly focuses on the static data PPP solution and accuracy analysis, and research on real-time dynamic PPP based on PPP-B2b services is scarce. Only one paper has been published on the PPP-B2b/INS loosely coupled integration (Xu et al., 2022), however, the loosely coupled integration fails to effectively solve the problem of significant decline in PPP positioning accuracy caused by signal interruption in dense urban environments. Moreover, research on the tightly coupled integration positioning of PPP-B2b/INS in dense environments has not been conducted. This study attempts to combine PPP-B2b with INS, con-

struct a real-time PPP-B2b/INS tightly coupled integration model based on PPP-B2b service, conduct vehicle-mounted GNSS tests in Zhengzhou, and conduct a PPP-B2b/INS tightly coupled integration positioning performance analysis under different urban environments. This study is of great significance for expanding the precision positioning applications of the Beidou PPP-B2b service.

2. PPP-B2b/INS tightly coupled integration model

2.1. PPP-B2b correction algorithm

The clock-offset correction and orbit correction broadcast by PPP-B2b only need to correct the clock error and satellite orbit coordinates calculated from broadcast ephemeris parameters in CNAV1 messages corresponding to BDS and LNAV messages corresponding to GPS (CSNO, 2020). The differential code bias was directly corrected to the pseudo-range observations.

Orbit correction parameters include the orbit correction vector $\delta O = [\delta O_{radial} \ \delta O_{along} \ \delta O_{cross}]$ components in the radial, track, and cross directions. The corrected satellite orbital position is calculated by the satellite position vector $(X_{broadcast}, Y_{broadcast}, Z_{broadcast})$ calculated according to the broadcast ephemeris parameters, and can be written as follows:

$$\begin{bmatrix} X_{orbit} \\ Y_{orbit} \\ Z_{orbit} \end{bmatrix} = \begin{bmatrix} X_{broadcast} \\ Y_{broadcast} \\ Z_{broadcast} \end{bmatrix} - \begin{bmatrix} \delta X \\ \delta Y \\ \delta Z \end{bmatrix} \quad (1)$$

In Eq. (1), $(X_{orbit}, Y_{orbit}, Z_{orbit})$ is satellite orbital position after correction, $(X_{broadcast}, Y_{broadcast}, Z_{broadcast})$ is satellite orbital position calculated from the broadcast ephemeris parameters, $(\delta X, \delta Y, \delta Z)$ is orbit correction broadcasted by PPP-B2b service, these can be calculated as follows:

$$e_{radial} = \frac{r}{|r|} \quad (2)$$

$$e_{cross} = \frac{r \times \dot{r}}{|r \times \dot{r}|} \quad (3)$$

$$e_{along} = e_{cross} \times e_{radial} \quad (4)$$

$$\begin{bmatrix} \delta X \\ \delta Y \\ \delta Z \end{bmatrix} = [e_{radial} \ e_{along} \ e_{cross}] \cdot \delta O^T \quad (5)$$

In Eqs. (2) and (3), r and $\dot{r}$ are satellite orbit position vectors and velocity vectors calculated from the broadcast ephemeris, respectively, and in Eqs. (4) and (5), e_{radial} , e_{along} , and e_{cross} are unit vectors in the radial, track, and cross directions, respectively.

Clock offset correction is a correction parameter for clock offset calculated by the broadcast ephemeris, and can be written as follows:

$$t_{satellite} = t_{broadcast} - \frac{C_0}{C} \quad (6)$$

In Eq. (6), $t_{broadcast}$ is the clock offset calculated from the broadcast ephemeris, $t_{satellite}$ is the corrected clock offset, C_0 is the correction parameter broadcast by the PPP-B2b service, and C is the velocity of light.

Differential code bias correction is used to eliminate the intersymbol deviation of each frequency, which can be written as follows:

$$\tilde{l}_{sig} = l_{sig} - DCB_{sig} \quad (7)$$

In Eq. (7), $\tilde{l}_{sig}$ is corrected pseudo range observation values, l_{sig} is original pseudo range observation values, DCB_{sig} is differential code bias of each frequency broadcasted by PPP-B2b service. The BDS-3 satellite must use the correction of the differential code bias to eliminate the intersymbol deviation of each frequency. Because the GPS broadcast ephemeris' clock offset imply the DCB of ionosphere-free combination, which eliminates the need for external input of inter symbol deviations. While BDS-3 satellites' clock errors are based on B3I, the DCB requires external input when different signal groups are used for ionosphere-free combination.

The parameter user range accuracy index ($URAI$) of each satellite is six bits; the highest three bits are defined as the user range accuracy class ($URAC_{CLASS}$), and the lowest three bits are defined as the user range accuracy value ($URAC_{VALE}$). The calculation formula for URA is as follows.

$$URA[mm] \leq 3^{URAC_{CLASS}} (1 + 0.25 * URAC_{VALE}) - 1 \quad (8)$$

2.2. Real-time PPP data processing model based on PPP-B2b correction parameters

The pseudo range and carrier phase observation equations are usually expressed as follows (Chen et al., 2016):

$$\begin{cases} P_i = \rho + c \cdot (dt_r - dt^s) + \delta Trop + \frac{f_i^2}{f_i^2 - f_j^2} \cdot I_1 + B_{r,i} - B_i^s + \varepsilon_{P_i} \\ L_i = \rho + c \cdot (dt_r - dt^s) + \delta Trop - \frac{f_i^2}{f_i^2 - f_j^2} \cdot I_1 + \lambda_i (N_i + b_{r,i} - b_i^s) + \varepsilon_{L_i} \end{cases} \quad (9)$$

In Eq. (9), i is the i frequency, P_i and L_i are pseudo range and phase observations, ρ is the geometric distance from satellite to receiver, dt_r and dt^s are clock offset inside receiver and satellite, f_i is the centre frequency point value of the i frequency, $\delta Trop$ is tropospheric delay, I_1 is the Ionospheric delay of L_1 , $B_{r,i}$ and B_i^s are code bias of receiver and satellite, $b_{r,i}$ and b_i^s are phase difference of receiver and satellite, ε_{P_i} and ε_{L_i} are unmodeled error in pseudo range measurement and carrier phase measurement. The relativistic effect, Sagnac delay, and phase winding are corrected according to the corresponding model (Gao and Shen, 2002).

An ionosphere-free model is typically used to eliminate the first-order term of ionospheric delay in dual-frequency PPP data processing. The combined equations

of the pseudo-range and carrier phase are as follows (Yao et al., 2019):

$$\begin{cases} P_{IF} = \alpha \cdot P_i + (1 - \alpha) \cdot P_j = \rho + c \cdot \bar{dt}_r - c \cdot dt_s + \delta Trop_{i,j} - B_{IF}^s + \varepsilon_{P_{IF}}^s \\ L_{IF} = \alpha \cdot L_i + (1 - \alpha) \cdot L_j = \rho + c \cdot \bar{dt}_r - c \cdot dt_s + \delta Trop_{i,j} + \lambda_{IF} \cdot \bar{N}_{IF} + \varepsilon_{L_{IF}}^s \end{cases} \quad (10)$$

$$\alpha = f_i^2 / (f_i^2 - f_j^2) \quad (11)$$

$$\lambda_{IF} \cdot \bar{N}_{IF} = \lambda_{IF} N_{IF} - b_{IF}^s + b_{r,IF} - B_{r,IF} \quad (12)$$

In Eqs. (10), (11) and (12), b_{IF}^s and $b_{r,IF}$ are phase difference of satellite and receiver in the ionosphere free model, B_{IF}^s and $B_{r,IF}$ are code bias of satellite and receiver in the ionosphere free model, λ_{IF} and N_{IF} are phase wavelength and ambiguity in the ionosphere free model, $\bar{dt}_r$ is the clock offset of receivers including code bias in the ionosphere free model, $\varepsilon_{P_{IF}}^s$ and $\varepsilon_{L_{IF}}^s$ are unmodeled error of pseudo range and carrier phase in the ionosphere free model; the code bias of ionosphere free combination at satellite can be corrected by the correction broadcast by PPP-B2b service. Due to the BDS-3 broadcast clock offset uses BDS-3 B3I as frequency reference, the satellite code hardware delay of B3I should be contained in the correction (Zhang et al., 2019). $B_{IF}^s - B_{B3I}^s$ will be the DCB between the ionosphere-free combination and reference frequency B3I at satellite, and the correction algorithm formula is as follows (Wang et al., 2019):

$$\begin{aligned} B_{IF}^s - B_{B3I}^s &= \left(\frac{f_i^2}{f_i^2 - f_j^2} B_i^s - \frac{f_j^2}{f_j^2 - f_i^2} B_j^s \right) - B_{B3I}^s \\ &= \frac{f_i^2}{f_i^2 - f_j^2} (B_i^s - B_{B3I}^s) - \frac{f_j^2}{f_j^2 - f_i^2} (B_j^s - B_{B3I}^s) \\ &= \frac{f_i^2}{f_i^2 - f_j^2} \tilde{l}_{i,B3I} - \frac{f_j^2}{f_j^2 - f_i^2} \tilde{l}_{j,B3I} \end{aligned} \quad (13)$$

In Eq. (13), $\tilde{l}_{i,B3I}$ and $\tilde{l}_{j,B3I}$ are pseudo-range observations of the i frequency and j frequency after correction by DCB corrections broadcast by the PPP-B2b service.

The extended Kalman filter is usually used to estimate unknown parameters; therefore, the variance of ionosphere-free combination observations is calculated as follows:

$$\sigma_k^2 = \sigma_{eph,k}^2 + \sigma_{noise,k}^2 \quad (14)$$

$$\sigma_{eph,k}^2 = (URA/1000)^2 \quad (15)$$

In Eqs. (14) and (15), k is the k th ionosphere-free combination observation, eph is the observation epoch, $\sigma_{eph,k}^2$ is the variance of satellite orbit and clock offset calculated by $URAI$, and $\sigma_{noise,k}^2$ is the variance of noise and multipath effect calculated by the elevation-weighted model. The

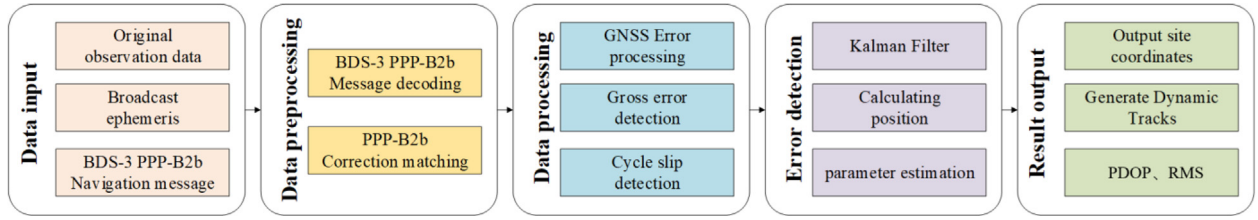


Fig. 1. Real-time ppp data processing model based on PPP-B2b correction parameters.

specific process of the real-time PPP data processing model based on the PPP-B2b correction parameters is shown in Fig. 1.

2.3. PPP-B2b/INS tightly coupled integration based on PPP-B2b correction parameters

The PPP/INS tightly coupled integration algorithm based on an ionosphere-free combination is a high-precision integrated navigation algorithm proposed after RTK/INS loosely/tightly coupled integration. This algorithm has a small dimension of the state vector and few observation equations, which makes the mathematical model easy to realise and requires fewer calculations. The PPP-B2b/INS tightly coupled integration based on PPP-B2b correction parameters has some differences with traditional PPP/INS tightly coupled integration model: Precise ephemeris are often used to predict position in traditional models, but after using broadcast ephemeris with PPP-B2b corrections to predict position in PPP-B2b/INS model, it can implement real-time positioning. The state vector (Abd Rabbou and El-Rabbany, 2014; Wang et al., 2017; Zhang et al., 2018) of the PPP-B2b/INS tightly coupled integration model is expressed as follows:

$$\mathbf{X}_k = [\delta P_{INS}^n \quad \delta v_{INS}^n \quad \delta \psi \quad \delta B_a \quad \delta B_g \quad \delta S_a \quad \delta S_g \quad \delta t_r \quad \delta i_r \quad \delta d_{wet} \quad \delta N_{IF}] \quad (16)$$

In Eq. (16), $\mathbf{X}_k$ is state vector of the k th epoch, δP_{INS}^n is the INS position in navigation coordinate system, δv_{INS}^n is the velocity in navigation coordinate system, $\delta \psi$ is the attitude correction, δB_a and δB_g are zero deviation error correction of accelerometer and gyroscope, δS_a and δS_g are scale factor error correction of accelerometer and gyroscope, δt_r and δi_r are clock error and clock drift of receiver, δd_{wet} is the tropospheric wet delay component, δN_{IF} is the ambiguity of ionosphere-free combination (Nassar and Elsheimy, 2006). The discrete system equation and observation equation of the PPP-B2b/INS tightly coupled integration model are as follows:

$$\mathbf{X}_k = \Phi_{k,k-1} \mathbf{X}_{k-1} + \omega_{k-1}, \omega_{k-1} \sim N(0, \mathbf{Q}_{k-1}) \quad (17)$$

$$\mathbf{Z}_k = \mathbf{H}_k \mathbf{X}_k + \eta_k, \eta_k \sim N(0, \mathbf{R}_{k-1}) \quad (18)$$

In Eqs. (17) and (18), $\mathbf{X}_{k-1}$ is the state vector of the $k-1$ th epoch, $\mathbf{Z}_k$ is a new observation vector of the k th epoch, $\mathbf{Q}_{k-1}$, and $\mathbf{R}_{k-1}$ are the prior variance of state noise and prior variance of observation noise in the $k-1$ th epoch, ω_{k-1} is the state noise of the $k-1$ th epoch, η_k

epoch, $k-1$ th epoch, design coefficient matrix $\mathbf{H}_k$, and state transition matrix $\Phi_{k,k-1}$ of dual-frequency ionosphere-free combination in the PPP-B2b/INS tightly coupled integration are as follows:

$$\mathbf{H}_k = [\mathbf{H}_{P_{IF}} \quad \mathbf{H}_{L_{IF}} \quad \mathbf{H}_{\dot{P}_{IF}}]^T \quad (19)$$

$$\Phi_{k,k-1} = \begin{bmatrix} \Phi_1 & \Phi_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \Phi_3 & \Phi_4 & \Phi_5 & \Phi_6 & 0 & \Phi_7 & 0 & 0 & 0 \\ 0 & 0 & \Phi_8 & 0 & \Phi_9 & 0 & \Phi_{10} & 0 & 0 \\ 0 & 0 & 0 & \Phi_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Phi_{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \Phi_{13} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \Phi_{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \Phi_{19} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \Phi_{18} \end{bmatrix} \quad (20)$$

In Eqs. (19) and (20), $\mathbf{H}_{P_{IF}}$, $\mathbf{H}_{L_{IF}}$, and $\mathbf{H}_{\dot{P}_{IF}}$ are design coefficients matrix for pseudorange, carrier, and Doppler observations in dual-frequency ionosphere-free combination; $\Phi_1, \dots, \Phi_{10}$ are state coefficients for position, velocity, and attitude in the PSI angular error model (Donald and Benson, 1975); $\Phi_{11}, \dots, \Phi_{14}$ are state coefficients for the zero deviation and scale factor of the IMU in the first-

order Gaussian Markov process; $\Phi_{19} = \begin{bmatrix} 1 & \Delta t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is

state coefficients for clock error, clock drift of receiver and dynamic equation of tropospheric wet delay, Φ_{18} is state coefficients for the ambiguity of ionosphere-free combination (Gao et al., 2017).

According to the discrete system equation and observation equation of the PPP-B2b/INS coupled integration model, the Kalman filtering time is updated as follows:

$$\mathbf{X}_{k,k-1} = \Phi_{k,k-1} \mathbf{X}_{k-1} \quad (21)$$

$$\mathbf{P}_k = \Phi_{k,k-1} \mathbf{P}_{k-1} \Phi_{k,k-1}^T + \mathbf{Q}_{k-1} \quad (22)$$

In Eqs. (21) and (22), $\mathbf{P}_{k-1}$ and $\mathbf{P}_k$ are the variance covariance matrices of the state parameters in the $k-1$ th epoch and the k th epoch. The specific structure of the PPP-B2b/INS tightly coupled integration model is shown in Fig. 2.

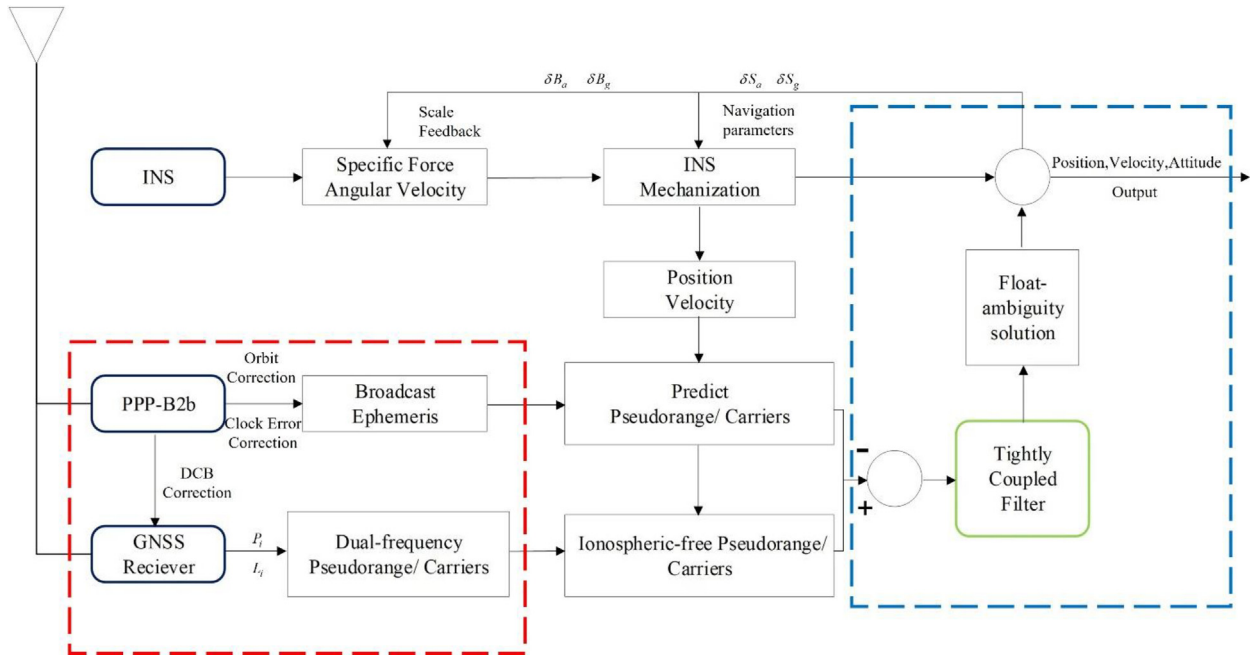


Fig. 2. PPP-B2b/INS tightly coupled integration model.

3. Positioning performance analysis of PPP-B2b/INS tightly coupled integration

3.1. Test equipment and data processing method

Two sections of sports car tests were conducted on 2 July 2022 in Zhengzhou City. Experiment 1 was conducted in an open-sky environment with low levels of building interference, and Experiment 2 was conducted in a more dense urban environment. Compared to Experiment 1, the satellite signal in Experiment 2 is more affected, satellite signals were interrupted several times due to the impact of complex environments. Table 1 lists the technical parameters of GNSS antennas, Table 2 lists the nominal parameters of inertial navigation system (XWYD7960). Two GNSS antennas which installed in the car forward and backward are connecting to INS, and these are used as calculating positioning reference results. The third GNSS antenna was connected to the special PPP-B2b GNSS receiver Sino M300 Pro, which was used to receive PPP-B2b messages and GNSS original observations. The sampling interval for the GNSS observation data was 1 s. Experiment 1 and Experiment 2 are all focusing on the

Table 1
Technical parameters of GNSS antennas.

Parameters	Precision
Pseudorange	BDS-3: B1 10 cm (1 σ), B2 10 cm (1 σ), B3 10 cm (1 σ)
Carrier phase	BDS-3: B1 1 mm (1 σ), B2 1 mm (1 σ), B3 1 mm (1 σ)
Timing	20 ns
Data quality	24-hour integrity rate $\geq 98\%$ 24-hour cycle skip ratio ≥ 8000 24-hour average multipath error < 0.3 m

Table 2

Nominal parameters of Inertial navigation system (XWYD7960).

Sensor	Gyroscope	Accelerometer
Bias	1°/h	0.5 mg
Scale factor	1500 ppm	4000 ppm
Random walk	0.005°/h	2 μ g/ $\sqrt{\text{Hz}}$

positioning performance of BDS satellites supported by PPP-B2b service. The installation of the experimental instruments is shown in Fig. 3(a) shows the top view of on-board test equipment, with GNSS antennas and INS stably placed on the roof; GNSS receiver M300 Pro is shown on the left of (b), on the right of (b) is the inertial navigation equipment XWYD796. The lever arm errors in three directions are 11.5/9.5/3 cm.

A GNSS base station was set up in the experiment, and the distance between the mobile station and the base station was maintained within 20 km. The smoothed RTK/INS tightly coupled integration results calculated by the commercial software Inertial Explorer (IE) released by NovAtel were used as a reference for PPP-B2b/INS tightly coupled integration positioning results. The specific processing method of PPP-B2b/INS tightly coupled integration positioning model based on the collected observation data and the correction information from PPP-B2b service are shown in Table 3 (Capitaine et al., 2000).

3.2. Precision analysis of PPP-B2b/INS tightly coupled integration positioning in open-sky environment

In Experiment 1, the moving track of the vehicle test equipped with the test equipment is shown in Fig. 4. The



Fig. 3. On-board test equipment.

Table 3
Data processing strategy of PPP-B2b/INS tightly coupled integration positioning model.

Processing Content	Processing Method
PPP model	Free ionospheric pseudorange/carrier phase combination
GNSS sampling interval	1 s
Observation cut-off altitude angle	5°
Parameter estimation	Kalman Filter
Satellite orbit correction	B1-CNAV1 broadcast ephemeris + PPP-B2b orbit correction
Satellite clock offset correction	B1-CNAV1 broadcast ephemeris + PPP-B2b clock offset correction
Ionospheric delay	Elimination of ionospheric first-order term in free ionospheric combination
Tidal effect	Correction according to the International Convention of IERS 2010
Phase winding/relativistic effect	Corresponding model correction
Tropospheric delay	Dry component corrected by Saastamoinen model + wet component corrected by parameter estimation
Ambiguity	Constant, floating solution

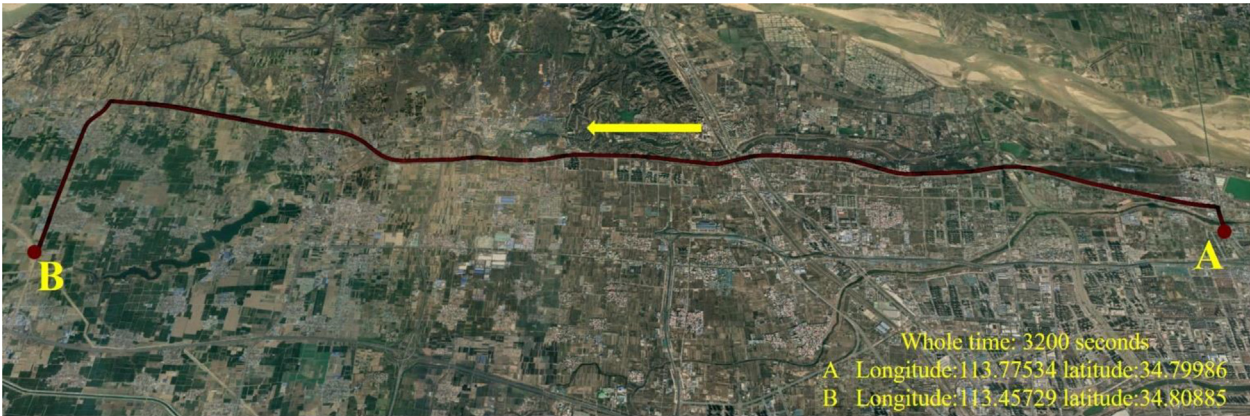


Fig. 4. Moving track of vehicle test in experiment 1.

starting point is at A, the ending point is at B, and the direction of car movement is from east to west, the duration of the experiment is 3200 s. The vehicle moved at a speed of less than 20 m/s in the east and north directions, and less than 5 m/s in the vertical direction to ensure the continuity of the number of receiving BDS satellites. The number of visible satellites and Position Dilution Of Precision (PDOP) of BDS are shown in Fig. 5. The number of visible satellites was kept within 10–18, with an average

of 14.7. The PDOP value is kept within 0.9–3.1, with an average of 1.19.

Fig. 6(a)–(c) respectively shows the positioning errors of PPP-B2b (BDS only), PPP-B2b (GPS only), PPP-B2b/INS and PPP/INS in three directions in an open-sky environment, respectively. The PPP-B2b and PPP-B2b/INS tightly coupled integration positioning experiments take the calculation results of the IE software as the reference value, and the results show that the positioning results are consistent

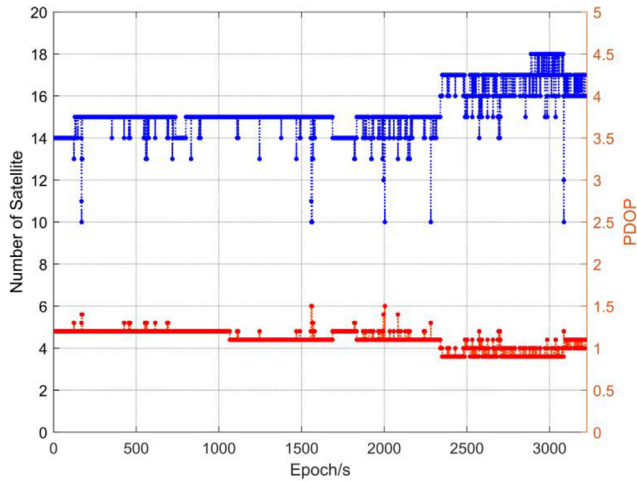
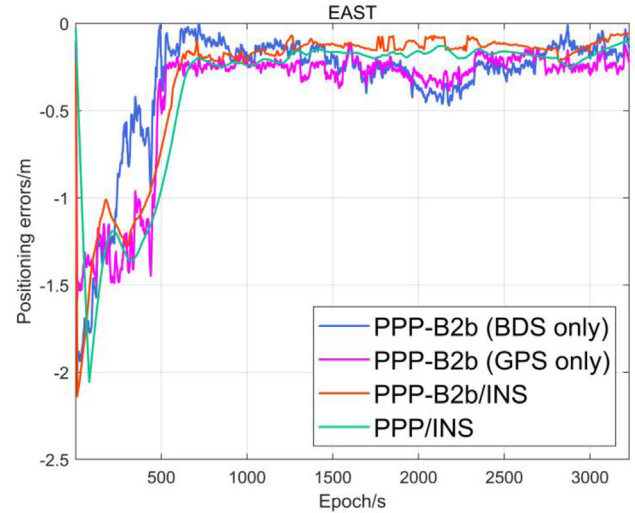


Fig. 5. The number of visible BDS satellites and PDOP in experiment 1.

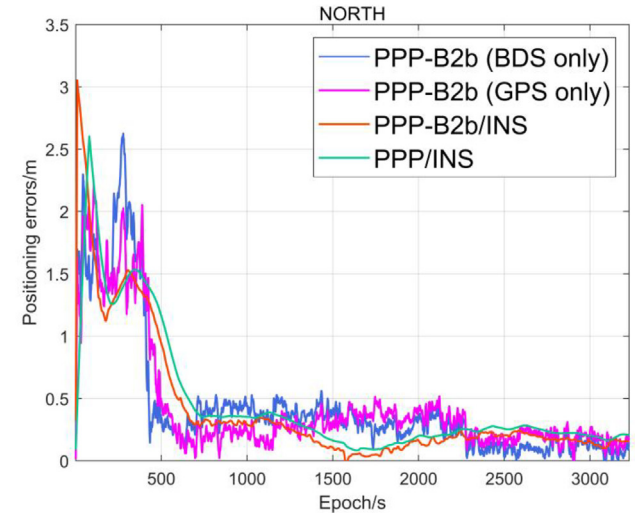
with the reference results. The positioning results were consistent and could achieve complete convergence. It can be seen that the real-time kinematic PPP-B2b can quickly converge to decimeter-level positioning accuracy for both BDS only and GPS only. And almost all positioning errors in the East, North and Up components are within 40 cm when the positioning errors tend to be stable, it means that the real-time kinematic PPP-B2b can achieve a positioning accuracy of several decimeters in open-sky environment. Fig. 6 also shows that the positioning errors of PPP-B2b/INS and PPP/INS tightly coupled integration only using BDS satellites, it can be seen that the convergence process is faster and the accuracy is higher by comparing with PPP-B2b, and the positioning errors in the East, North and Up components are within 20 cm after it converge.

Convergence time is generally defined as meeting the accuracy requirements and maintaining the accuracy status for at least 5 min. The data obtained from the experiment are based on the standard in the Beidou Satellite Navigation System Space Signal Interface Control Document Precision Single Point Positioning Service Signal PPP-B2b (Version 1.0) released by the China Satellite Navigation System Management Office in July 2020. The horizontal positioning accuracy is not more than 30 cm. The positioning accuracy in the elevation direction should not exceed 60 cm as the precision single point positioning convergence accuracy standard, and the convergence time to reach the specified accuracy is counted. The convergence results and positioning errors in Table 4 were obtained from the statistics of the experimental positioning results, in which the positioning error was evaluated by the root mean square errors (RMS).

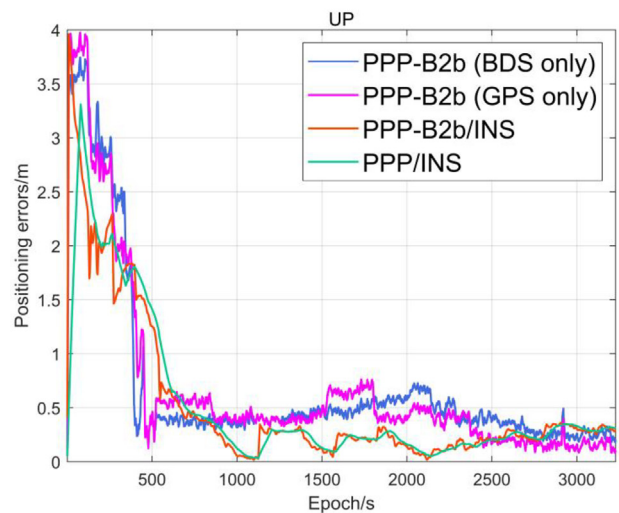
To demonstrate the positioning performance of the real-time PPP with PPP-B2b service, we not only calculate the positioning errors of PPP-B2b (BDS only) and PPP-B2b (GPS only), but also evaluate the real-time PPP using broadcast ephemeris (abbreviated as PPP-Broadcast) and



(a) Positioning errors in the east direction



(b) Positioning errors in the north direction



(c) Positioning errors in the up direction

Fig. 6. Positioning error of PPP-B2b (BDS only), PPP-B2b (GPS only), PPP-B2b/INS, PPP/INS in open-sky environment.

Table 4

Positioning error and convergence time Of PPP-B2b and PPP-B2b/INS tightly coupled integration in open-sky environment.

Solution mode	RMSE (cm)			Convergence time (s)
	East	North	Up	
PPP-Broadcast (BDS only)	57.6	48.8	78.9	1068
PPP-B2b (BDS only)	31.7	21.3	39.6	843
PPP-B2b (GPS only)	34.1	27.6	41.0	856
PPP-Precise (BDS only)	38.9	31.0	42.3	852
PPP-B2b/INS (BDS only)	17.2	10.3	19.4	773
PPP/INS (BDS only)	21.6	17.3	23.7	787

the real-time PPP using precise ephemeris (abbreviated as PPP-Precise) for comparison. Table 4 lists the corresponding statistical results of positioning error and convergence time for PPP, PPP-B2b, PPP-B2b/INS tightly coupled integration and traditional PPP/INS tightly coupled integration. And the following conclusions also can be drawn from Table 4:

- (1) The positioning errors of real-time kinematic PPP and PPP-B2b/INS tightly coupled integration using PPP-B2b service is slightly better than that of real-time PPP-Precise and PPP/INS which using corrections provided by CNES-RT. The reason may be that we use CNES-RT to calculate PPP-Precise and PPP/INS, but only a limited number of BDS-3 satellites have CNES-RT orbits and clock offsets, it will affect the positioning performance of PPP-Precise.
- (2) The positioning errors of real-time kinematic PPP using PPP-B2b service is significantly better than that of PPP-Broadcast. However, the positioning accuracy of PPP-Precise still can achieve decimeters, it means that the accuracy of BDS-3 broadcast ephemeris can meet high standards.
- (3) As shown in Fig. 6, the positioning error of PPP-B2b (GPS only) is similar to PPP-B2b (BDS only). The RMSE of PPP-Broadcast and PPP-Precise is respectively 57.6/48.8/78.9 and 38.9/31.0/42.3 cm. The RMSE of PPP-B2b were 31.7/21.3/39.6 cm in the east, north and up directions, respectively. It indicating that real-time kinematic PPP using the corrections provided by PPP-B2b service for GPS and BDS can achieve dynamic decimeter-level.
- (4) The positioning accuracy of the tightly coupled integration of PPP-B2b/INS was improved compared to that of PPP-B2b and PPP/INS. The RMSE in the east, north, and up directions were 17.2, 10.3/19.4 cm, which increased by 39.2%, 51.6%, and 51.0%, respectively. The convergence time of the PPP-B2b/INS tightly coupled integration was 70 s faster than that of PPP-B2b.
- (5) By comparing PPP-B2b and PPP-Precise, PPP-B2b/INS and PPP/INS, the positioning accuracy of real-time PPP-Precise is lower than PPP-B2b and the positioning accuracy of real-time PPP-INS is lower than

PPP-B2b/INS, it means BDS-3 PPP-B2b service is better than real-time precise ephemeris provided by CNES-RT when calculating real-time kinematic PPP.

It is recognised that PPP-B2b is unable to obtain a high-precision solution during GNSS signal outages, whereas PPP-B2b/INS tightly coupled integration continues to work by INS mechanisation. In the data of Experiment 1, we conducted four periods of GNSS signal outage experiments to test the superiority of PPP-B2b/INS tightly coupled integration and block the GNSS signal for 30 s at the 1000th, 1500th, 2000th, and 2500th epochs. It can be observed from Fig. 7 that the error gradually increases with an increase in the GNSS signal interruption time. Table 5 shows the positioning error change in the four-segment experiment. The RMS error within 30 s of GNSS signal interruption in the four segments are 33.7/18.3/24.6 cm in the east, north, and up directions, respectively. By amplifying four simulations in Fig. 7, these four enlarged images shows that the positioning differences between before and after simulations, it can be seen that with the support of high-precision INS, even if the GNSS signal is outage, INS can maintain high accuracy while working independently for periods of time, the positioning error of tightly coupled integration in open environments can still reach the meter level, and the PPP-B2b/INS tightly coupled integration can quickly converge after signal recovery.

3.3. Precision analysis of PPP-B2b/INS tightly coupled integration positioning in dense urban environment

Considering the complexity of the urban environment, there were many signals blocking the buildings in the motion track of Experiment 2. The equipment used in the experiment was the same as that used in Experiment 1. The moving track of the automobile experiment is shown in Fig. 8. The starting point and ending point are both at A, and the duration of the experiment is 3386 s. The number of visible satellites and position accuracy factor of BDS are shown in Fig. 9. The number of visible satellites was kept within 1–18, with an average of 9.7. The PDOP value was kept within 0–20, with an average of 3.1.

Experiment 2 lasted approximately 60 min, passed through three bridges, as shown in Fig. 10, and the GNSS

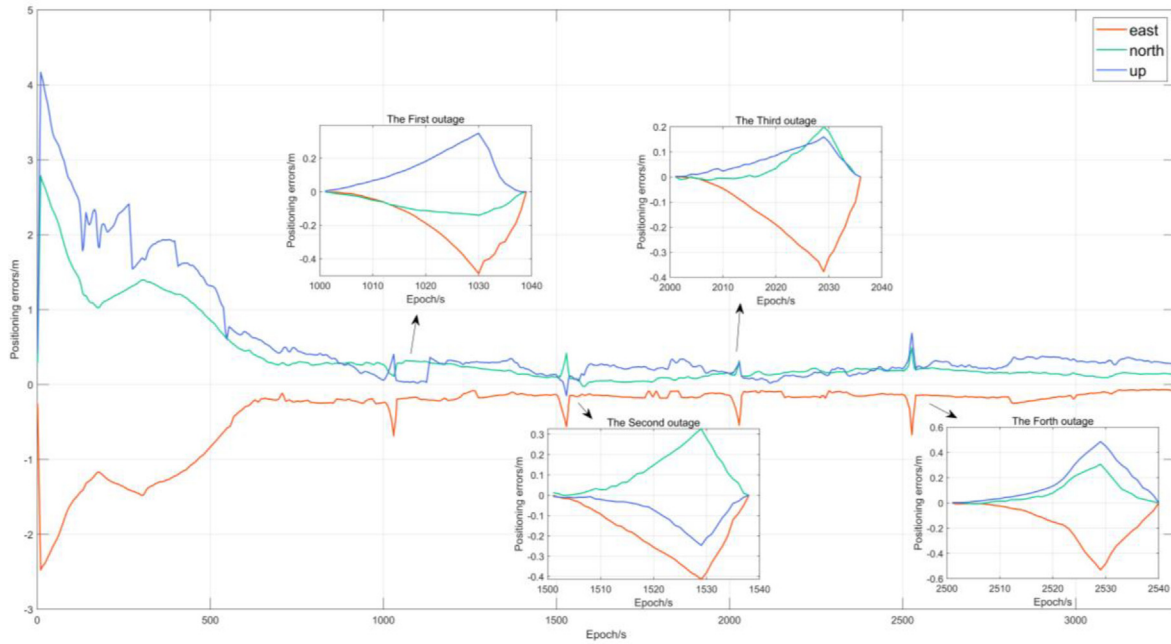


Fig. 7. Positioning error of PPP-B2b/INS tightly coupled integration during simulated GNSS signal outages.

Table 5
Statistics positioning errors for PPP-B2b/INS tightly coupled integration during gnss outages (Unit: cm).

Direction	5 s	10 s	15 s	20 s	25 s	30 s
East	0.7	4.4	9.8	18.4	25.1	33.7
North	0.8	2.3	5.1	8.5	13.9	18.3
Up	1.2	3.6	6.4	11.0	19.3	24.6

signal was blocked for 18 s, 9 s and 5 s respectively at (a), (b) and (c). During these three periods, the PPP-B2b positioning accuracy did not converge quickly because the number of satellites was less than four and could not meet

the requirements of satellite positioning. Therefore, after the GNSS signal is recovered, PPP-B2b converges. Fig. 11 shows the positioning error of PPP-B2b (BDS only and GPS only), and the RMSE of it in the east, north and up directions were 141.3/95.2/156.6 and 147.5/96.4/159.3 cm, respectively. Fig. 11 also shows the positioning error of PPP-B2b/INS and PPP/INS tightly coupled integration. The root mean square errors in the east, north, and up directions were 30.0/28.4/46.3 cm, respectively. It can be clearly seen that the accuracy of PPP-B2b/INS is obviously better than that of PPP-B2b and PPP/INS. After the signal is missing, using the feature that a tightly coupled integration of fewer than four satel-

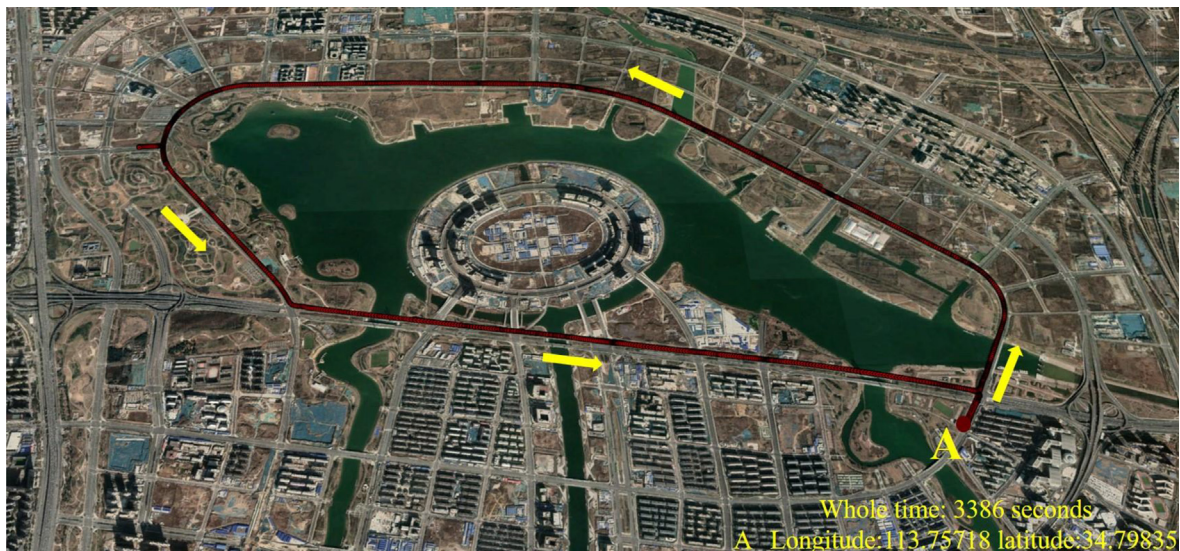


Fig. 8. Moving track of vehicle test in experiment 2.

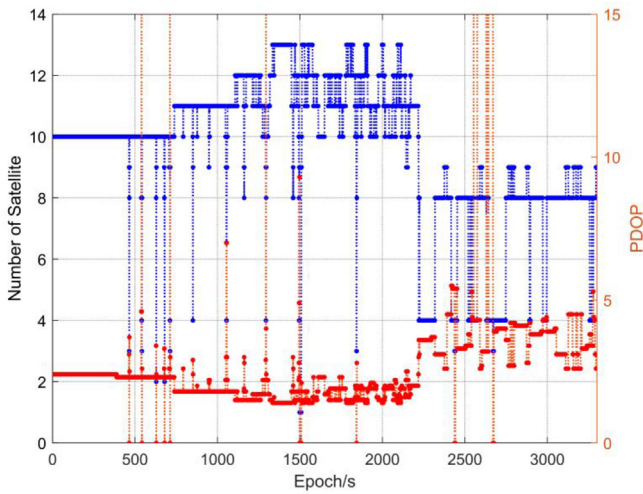


Fig. 9. The number of visible satellites and PDOP in experiment 2.

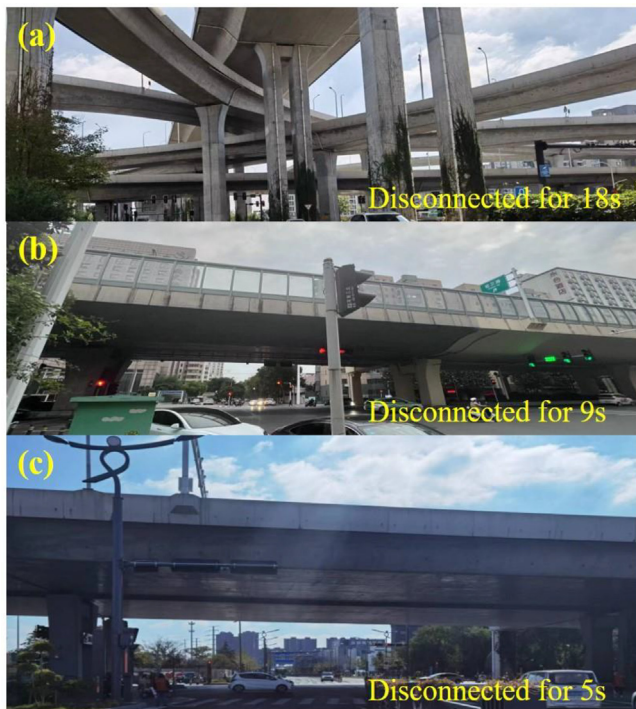
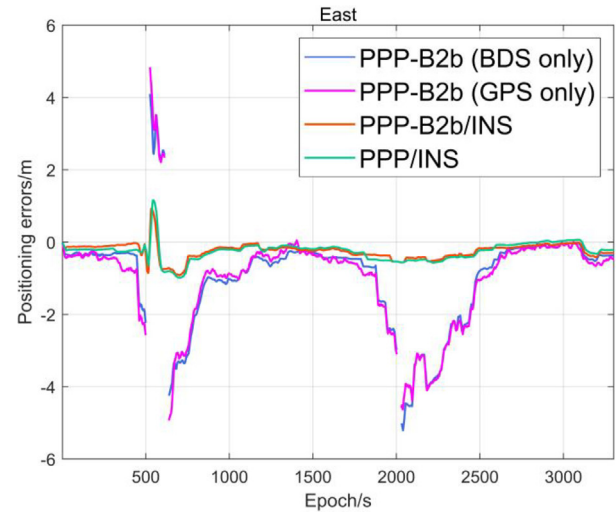


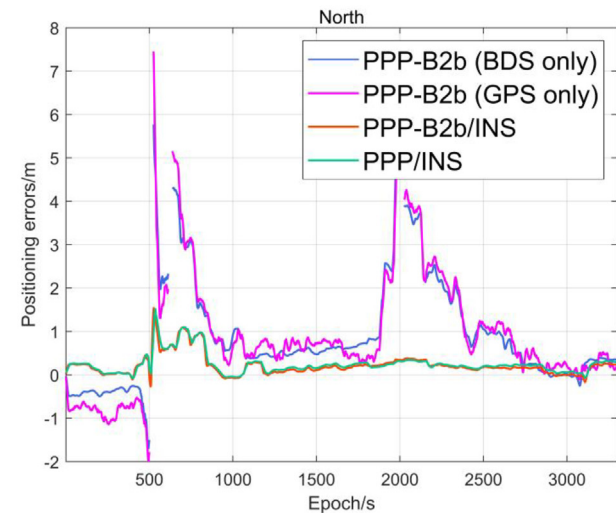
Fig. 10. Three bridges in experiment 2.

lites can still continue to position, higher positioning accuracy can still be obtained after satellite signal recovery. In a dense urban environment, when the GNSS signal is interrupted, PPP-B2b plays a small role, but tightly coupled integration with INS can effectively converge the error and greatly improve the positioning accuracy.

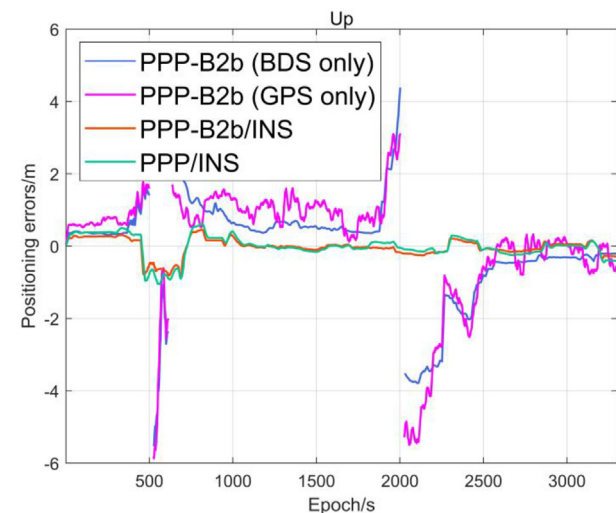
As Table 6 shows, on the one hand, it can be seen that real-time kinematic PPP using broadcast ephemeris cannot achieve decimeters level due to the signal blocked, although using broadcast ephemeris and PPP-B2b services can improve the positioning accuracy, but the positioning



(a) Positioning errors in the east direction



(b) Positioning errors in the north direction



(c) Positioning errors in the up direction

Fig. 11. Positioning error of PPP-B2b (BDS only), PPP-B2b (GPS only), PPP-B2b/INS, PPP/INS in dense urban environment.

Table 6

Positioning error and convergence time of PPP-B2b and PPP-B2b/INS tightly coupled integration in dense urban environment.

Solution mode	RMS (cm)		
	East	North	Up
PPP-Broadcast (BDS only)	183.4	146.5	201.3
PPP-B2b (BDS only)	141.3	95.2	156.6
PPP-B2b (GPS only)	147.5	96.4	159.3
PPP-Precise (BDS only)	151.2	99.7	163.5
PPP-B2b/INS (BDS only)	30.0	28.4	46.3
PPP/INS (BDS only)	41.7	36.1	53.4

accuracy cannot achieve decimeters level in three directions. On the other hand, the performance of the PPP-B2b/INS tightly coupled integration during signal outages was significantly improved compared to PPP-B2b in Experiment 2, with respective improvements in positioning accuracy in the east, north, and up components of 76.4%, 65.8%, and 69.1%, respectively, and the accuracy of the PPP-B2b/INS tightly coupled integration is slightly better than PPP/INS tightly coupled integration.

Due to three GNSS signal outages, the positioning of PPP-B2b was also interrupted three times in the 500th, 627th and 2000th epoch, as Fig. 11 showed. It can be seen that due to the short time interval between the first outage and the second outage, the second outage occurred before the reconvergence was completed after the first outage, which also caused some fluctuations in the positioning errors of PPP-B2b/INS tightly coupled integration during this period in Fig. 11, the sequence diagram of PPP-B2b/INS tightly coupled integration repeatedly bulged when these two outages occurred, which also affected the positioning error of the entire experiment to a certain extent. When the third outage occurs, the positioning error sequence diagram does not change significantly due to the short outage time. By comparing the positioning error showed in Fig. 11 with it in Fig. 7, the reconvergence can be achieved after GNSS signal recovery and achieve high accuracy, it means that tightly coupled integration have important implications for reconvergence both in open-sky or dense urban environments during GNSS outages. According to the standard in the Beidou Satellite Navigation System Space Signal Interface Control Document Precision Single Point Positioning Service Signal PPP-B2b (Version 1.0), the positioning errors of PPP-B2b and PPP-B2b/INS tightly coupled integration in Experiment 2 were statistically analyzed. The statistical results showed that even in dense urban environments, real-time PPP and PPP/INS based on PPP-B2b services can still achieve convergence and achieve decimeter level accuracy.

4. Conclusion

Aiming at the problem of reducing PPP positioning accuracy due to the unavailability of RTS when the Internet is interrupted, this study combines real-time high-

precision correction provided by PPP-B2b services with PPP/INS in different scenarios and draws the following conclusions:

In the open-sky environment, we conduct researches on real-time kinematic PPP in four forms. Dual-frequency real-time kinematic PPP based on PPP-B2b service can achieve decimeters level. The RMSE of PPP-B2b (BDS only and GPS only) in the east, north and up directions is 31.7/21.3/39.6 and 34.1/27.6/41.0 cm respectively, and the RMSE of PPP-B2b/INS tightly coupled integration in the east, north and up directions is 7.5/12.9/18.3 cm respectively. The precision of the PPP-B2b/INS tightly coupled integration was 39.2%, 51.6%, and 51.0% higher than that of PPP-B2b (BDS-only) in the three directions. On the other hand, the positioning accuracy of PPP-B2b is also higher than that of traditional PPP-Broadcast and PPP-Precise. The convergence time is also statisticed in open-sky environment, it can be seen that the real-time kinematic PPP using PPP-B2b correction can shorten convergence time compared with PPP-Broadcast and PPP-Precise.

In the dense urban environment, three bridges caused GNSS signal outages for 18 s, 9 s, and 5 s respectively. The root mean square errors of PPP-B2b (BDS only and GPS only) in the east, north, and up directions are 141.3, 95.2/156.6 and 147.5/96.4/159.3 cm respectively. The RMSE of the PPP-B2b/INS tightly coupled integration in the east, north, and up directions are 30.0/28.4/46.3 cm. The positioning results of PPP-B2b and traditional PPP (PPP-Broadcast and PPP-Precise) could not achieve the decimetre-level positioning accuracy in dense urban when GNSS signal blocked, but the precision of the PPP-B2b tightly coupled integration positioning was significantly improved. The precision of tightly coupled integration increased by 76.4%, 65.8%, and 69.1% in the three directions compared to PPP-B2b.

The experiments in this study were mainly based on dynamic vehicle experiments in an urban environment. The results of the experiments show that the positioning accuracy of the BDS-3 PPP-B2b service in an urban environment is high, and the real-time kinematic PPP using PPP-B2b corrections is not limited by the Internet. The corrections provided by PPP-B2b have significant effect to the real-time kinematic PPP when using GPS or BDS alone. However, when encountering a complex environment or many signal obstructions in a test, the number of visible satellites is insufficient, which reduces the positioning accuracy of the PPP-B2b service. Using PPP-B2b/INS tightly coupled integration can continuously obtain the positioning accuracy at the decimetre level, which is of great significance for high-precision navigation and positioning rather than traditional PPP/INS tightly coupled integration.

The simulation of GNSS signal outages in open-sky environments and the natural momentary outages in dense urban environments also showed that the PPP-B2b/INS tightly coupled integration can reconverge quickly. By comparing the positioning results in these two scenarios,

the following conclusions can be drawn that the tightly coupled integration can obtain high precision positioning result even if the number of satellites is less than 4, INS can efficiently locate independently when GNSS signal outages, so the tightly coupled integration can achieve fast convergence after signal recovery.

To explore the applicability of the algorithm at a deeper level, further work could be performed on the following aspects: ① Researching the multi-GNSS system based on PPP-B2b service broadcasted by SBAS; ② Exploring methods for obtaining fixed-ambiguity solutions for real-time kinematic PPP assisted by PPP B2b services; ③ Investigating filtering methods such as robust adaptive filtering to control error divergence when GNSS signal outages in PPP/INS tightly coupled integration positioning; ④ Considering the variability of the actual environment, additional experiments should be conducted in different places.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Abd Rabbou, M., El-Rabbany, A., 2014. Tightly coupled integration of GPS precise point positioning and MEMS-based inertial systems. *GPS Solut.* 19 (4), 601–609. <https://doi.org/10.1007/s10291-014-0415-3>.
- Angrisano, A., Gaglione, S., Gioia, C., 2013. Performance Assessment of GPS/GLONASS single point positioning in an urban environment. *Acta Geod. Geophys.* 48 (2), 149–161. <https://doi.org/10.1007/s40328-012-0010-4>.
- Capitaine, N., Guinot, B., McCarthy, D., 2000. Definition of the celestial ephemeris origin and of UT¹ in the International reference frame. *Astron* 355, 98–405.
- Chen, C., Chang, G., 2020. PPPLib: an open-source software for precise point positioning using GPS, BeiDou, Galileo, GLONASS, and QZSS with multi-frequency observations. *GPS Solut.* 25, 18. <https://doi.org/10.1007/s10291-020-01052-4>.
- Chen, J., Wang, J., Zhang, Y., et al., 2016. Modeling and assessment of GPS/BDS combined precise point positioning. *Sensors* 16, 1151. <https://doi.org/10.3390/s16071151>.
- Collins, P., Bisnath, S., Lahaye, F., et al., 2010. Undifferenced GPS ambiguity resolution using the decoupled clock model and ambiguity datum fixing. *Navigation* 57 (2), 123–135. <https://doi.org/10.1002/j.2161-4296.2010.tb01772.x>.
- CSNO, 2020. BeiDou navigation satellite system signal in space interface control document precision single point positioning service signal PPP-B2b (Version 1.0). China Satellite Navigation Office. Available online: <http://www.beidou.gov.cn/xt/gfzx/202008/P020200803362062482940.pdf> (Accessed on 3 August 2020).
- Donald, O., Benson, J.R., 1975. A comparison of two approaches to pure-inertial and doppler-inertial error analysis. *IEEE Trans. Aerosp. Electron. Syst.* 11 (4). <https://doi.org/10.1109/TAES.1975.308106>.
- Gao, Y., Shen, X., 2002. A new method for carrier-phase-based precise point positioning. *Navigation* 49 (2), 109–116. <https://doi.org/10.1002/j.2161-4296.2002.tb00260.x>.
- Gao, Z., Zhang, H., Ge, M., et al., 2015. Tightly coupled integration of ionosphere-constrained precise point positioning and inertial navigation systems. *Sensors* 15 (3), 5783–5802. <https://doi.org/10.3390/s150305783>.
- Gao, Z., Zhang, H., Ge, M., et al., 2017. Tightly coupled integration of multi-GNSS PPP and MEMS inertial measurement unit data. *GPS Solut.* 21 (2), 377–391. <https://doi.org/10.1007/s10291-016-0527-z>.
- Gong, Y., Cai, C., Dai, W., et al., 2020. Least-squares collocation modeling of regional ionospheric TEC for accelerating real-time single-frequency PPP convergence. *IET Radar Sonar Navig.* 13 (6), 1031–1038. <https://doi.org/10.1049/iet-rsn.2018.5504>.
- Gu, S., Dai, C., Fang, W., et al., 2021. Multi-GNSS PPP/INS tightly coupled integration with atmospheric augmentation and its application in urban vehicle navigation. *J. Geod.* 95 (6), 1–15. <https://doi.org/10.1007/s00190-021-01514-8>.
- Hadas, T., Bosy, J., 2015. IGS RTS precise orbits and clocks verification and quality degradation over time. *GPS Solut.* 19 (3), 93–105. <https://doi.org/10.1007/s10291-014-0369-5>.
- Héroux, P., Kouba, J., 2001. GPS precise point positioning using IGS orbit products. *Phys. Chem. Earth Part A* 26 (6–8), 573–578. [https://doi.org/10.1016/S1464-1895\(01\)00103-X](https://doi.org/10.1016/S1464-1895(01)00103-X).
- Li, T., Zhang, H., Niu, X., et al., 2017. Tightly-Coupled integration of Multi-GNSS single-frequency RTK and MEMS-IMU for enhanced positioning performance. *Sensors* 17, 2462. <https://doi.org/10.3390/s17112462>.
- Liu, Y., Yang, C., Zhang, M., 2022. Comprehensive analyses of PPP-B2b performance in China and surrounding areas. *Remote Sens.* 14, 643. <https://doi.org/10.3390/rs14030643>.
- Nassar, S., El-Sheimy, N., 2006. A combined algorithm of improving INS error modeling and sensor measurements for accurate INS/GPS navigation. *GPS Solut.* 10 (1), 29–39. <https://doi.org/10.1007/s10291-005-0149-3>.
- Nie, Z., Gao, Y., Wang, Z., et al., 2018. An approach to GPS clock prediction for real-time PPP during outages of RTS stream. *GPS Solut.* 22 (1), 1–14. <https://doi.org/10.1007/s10291-017-0681-y>.
- Ren, Z., Gong, H., Peng, J., et al., 2020. Performance assessment of real-time precise point positioning using BDS PPP-B2b service signal. *Adv. Space Res.* 68 (8), 3242–3254. <https://doi.org/10.1016/j.asr.2021.06.006>.
- Wang, H., Sun, F., Xiao, K., et al., 2017. Research progress and prospect of PPP/INS integration system. *Global Position. Syst.* 42 (5), 53–58. <https://doi.org/10.13442/j.gnss.1008-9268>.
- Wang, J., Yang, Y., Zhang, Q., et al., 2019. Analysis of inter-system bias in Multi-GNSS precise point positioning. *Geomat. Inf. Sci. Wuhan Univ.* 44 (4), 475–481. <https://doi.org/10.13203/j.whugis20170132>.
- Xu, X., Nie, Z., Wang, Z., Wang, B., et al., 2022. Performance assessment of BDS-3 PPP-B2b/INS loosely coupled integration. *Remote Sens.* 14 (13), 2957. <https://doi.org/10.3390/rs14132957>.
- Yang, Y., Ding, Q., Gao, W., et al., 2022. Principle and performance of BDSBAS and PPP-B2b of BDS-3. *Satell. Navig.* 3 (1), 5. <https://doi.org/10.1186/s43020-022-00066-2>.
- Yang, H., Ji, S., Weng, D., et al., 2021. Assessment of the feasibility of PPP-B2b service for real-time coseismic displacement retrieval. *Remote Sens.* 13, 5011. <https://doi.org/10.3390/rs13245011>.
- Yao, Y., Peng, W., Xu, C., et al., 2019. The realization and evaluation of mixed GPS/BDS PPP ambiguity resolution. *J. Geod.* 93 (9), 1283–1295. <https://doi.org/10.1007/s00190-019-01245-x>.
- Zhang, Y., Gao, Y., 2008. Integration of INS and un-differenced GPS measurements for precise position and attitude determination. *J. Navig.* 61 (1), 87–97. <https://doi.org/10.1017/S0373463307004432>.
- Zhang, R., He, Z., Ma, L., Xiao, G., et al., 2022. Analysis of BDS-3 PPP-B2b positioning and time transfer service. *Remote Sens.* 14, 2769. <https://doi.org/10.3390/rs14122769>.

- Zhang, Z., Wang, Y., Teng, H., et al., 2018. BDS/GPS integrated precise point positioning algorithm based on high-degree DMCKF. *Modern Defence Technol.* 46 (3), 48–53. <https://doi.org/10.3969/j.jssn.1009-086x>.
- Zhang, X., Zhu, F., Zhang, Y., et al., 2019. The improvement in integer ambiguity resolution with INS aiding for kinematic precise point positioning. *J. Geod.* 93 (7), 993–1010. <https://doi.org/10.1007/s00190-018-1222-3>.
- Zumberge, J.F., Heflin, M.B., Jefferson, D.C., et al., 1997. Precise point positioning for the efficient and robust analysis of GPS data from large networks. *J. Geophys. Res.* 102 (B3), 5005–5017. <https://doi.org/10.1029/96JB03860>.



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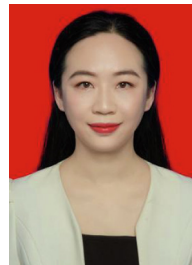
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